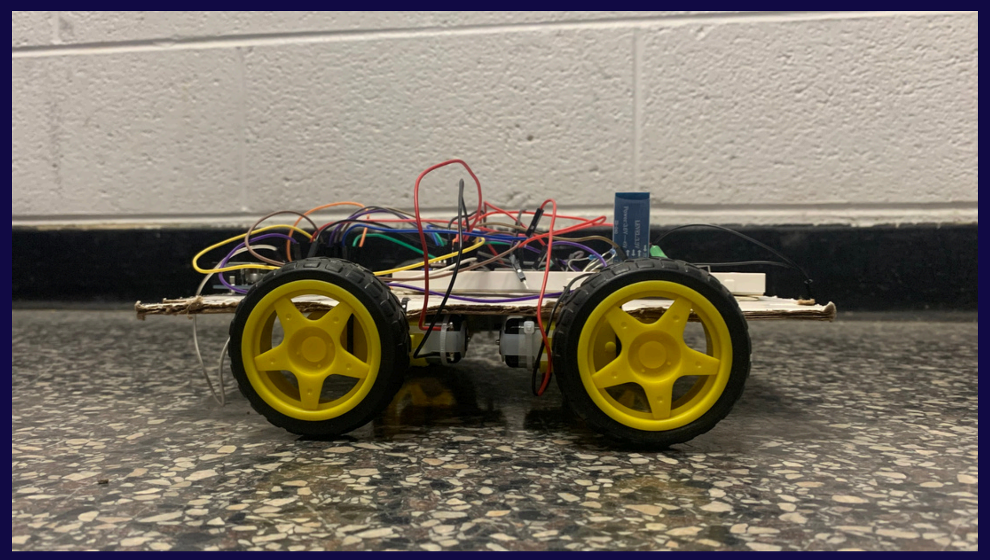
THE T-BOT

A GESTURE CONTROLLED CAR



An early prototype

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INTRODUCTION

For years, the exciting, fast growing field of Tiny Machine Learning (TinyML) has offered stunning new possibilities, and for my project, I decided that I wanted to explore this field. Over the course of a year, I have trained my own neural network to recognize simple gestures on a camera, compressed it to run on a microcontroller with extremely constrained resources, and deployed it to a working car.

OBJECTIVE

The objective of this project was to create a fully functioning system running entirely on a microcontroller with 256KB of RAM. The user would make specific gestures at a camera, which would relay the image data back to the brain. Then, the brain would perform image recognition to determine what gesture was performed, relay it to the car, which would move in the according direction.

SYSTEM OVERVIEW

Preprocessing: The brain receives the image from the camera and runs a number of preparation functions on it

Inference: The image is fed into the neural network, which yields which gesture it thinks it is most likely to be

Relay: The brain sends the command over serial to a different nano,

Movement: The wheels are spun accordingly to drive the car in the requested direction

STEP 1: THE CAR

I started out this project a novice in Arduino, and by the end, I was a master. After spending countless hours designing, thinking, modeling, and prototyping, I finally built the first iteration of the car.

Then, after building the car, I had to figure out how to power and control it. I tried out many different solutions before choosing lithium-ion batteries, with power lines of different voltage. Coding it to run with a joystick took lots of effort, as did properly attaching a bluetooth sensor to my module. Finally, after months of research, design, planning, I completed my car.

STEP 2: THE NEURAL NETWORK

After designing the car within the first half of the year, I had a large challenge for me in the second half of the year: not only training my own neural network, but fitting it onto the microcontroller with a reasonable runtime.

After spending researching solutions, I found that TinyML, a cutting edge technology developed in 2019, would let me slim down a traditional neural network, a collection of data typically around 12 MB, to 200 KB. This reduction of about 600 times in size is achieved through techniques like quantization, pruning, and more.

STEP 3: THE COMBINATION

Finally, I put everything together. Using the publicly available HaGRID dataset on Kaggle, I trained a neural network to recognize simple gestures, and fine tuned it with images I collected myself. Then, applying TinyML techniques, I shrunk the neural network down by a factor of 600, and deployed it to microcontroller. Lastly, I connected the camera module to the microcontroller and wrote the code to perform inference on-edge with minimal latency, and set up the resulting movement patterns up for the car. After a year of hard work, I had managed to make my own TinyML project.

METHODOLOGY

First, I built a cardboard chassis car with four TT motors that steered via differential speed control from a joystick. Then, I trained a TinyML neural network small enough to run on an Arduino Nano 33 BLE Sense. Finally, I linked everything together wirelessly via Bluetooth, replacing a joystick with hand gesture control.

CONCLUSION

This project demonstrated that machine learning doesn't require expensive, power-hungry hardware to be useful. By training a gesture recognition model and deploying it in a microcontroller with severely constrained resources, I've been convinced that the future of Machine Learning isn't in data centers, but the small, cheap devices surrounding us

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"THE FUTURE OF MACHINE LEARNING IS TINY AND BRIGHT"

PROFESSOR VIJAY JANAPA REDDI